

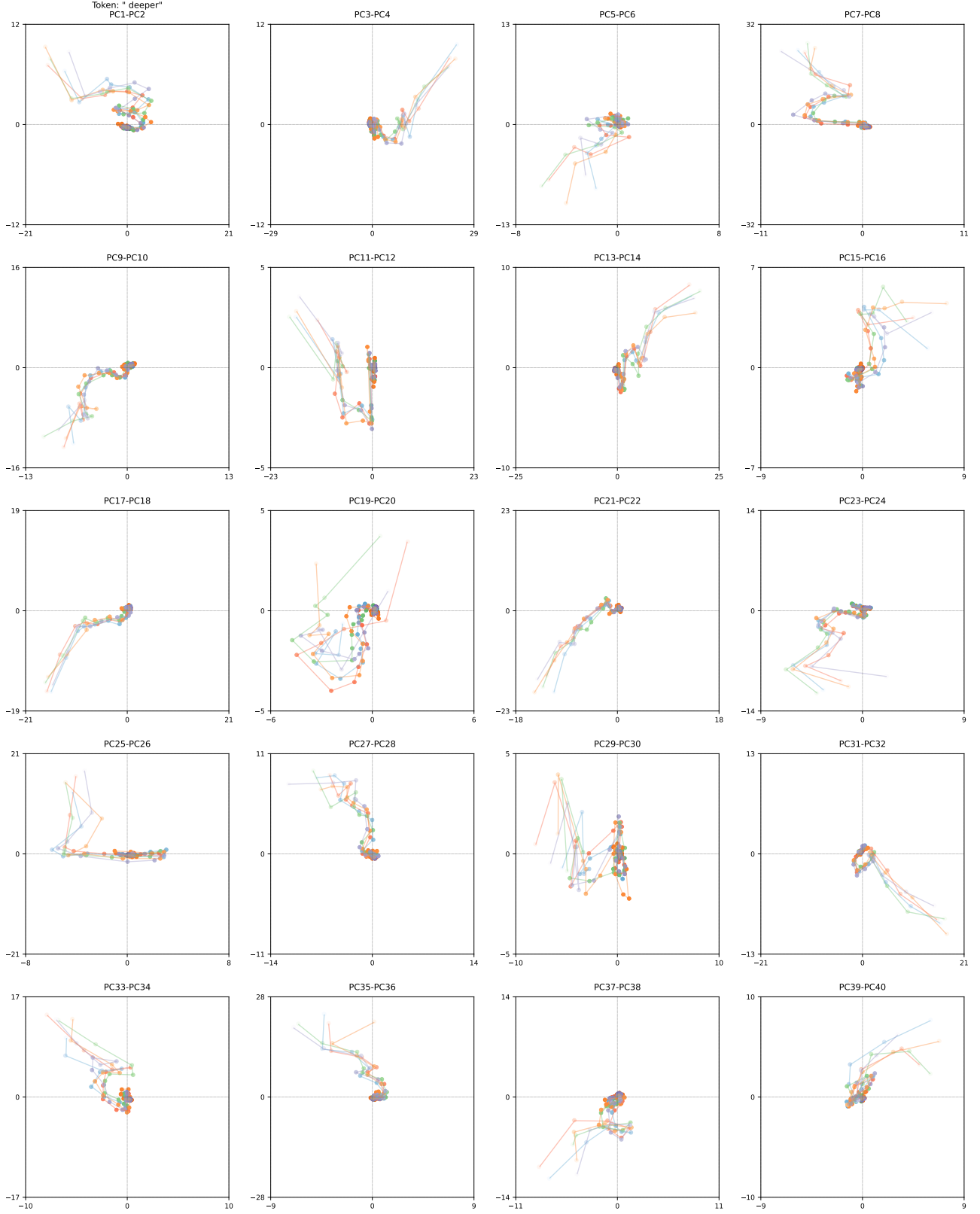


Figure 24: Detailed PCA of Latent Space trajectories for the trivia question. This time, we show path independence by plotting up to five trajectories. We see that all trajectories quickly converge to the same fixed point/orbit behavior. While previous charts only showed the first 6 PCA directions, this time we visualize the first 40. Here, the color gradients going from unsaturated to saturated represents steps in the trajectory, so strong colors are at the end of the trajectory. Gray denotes the overlap of multiple trajectories.

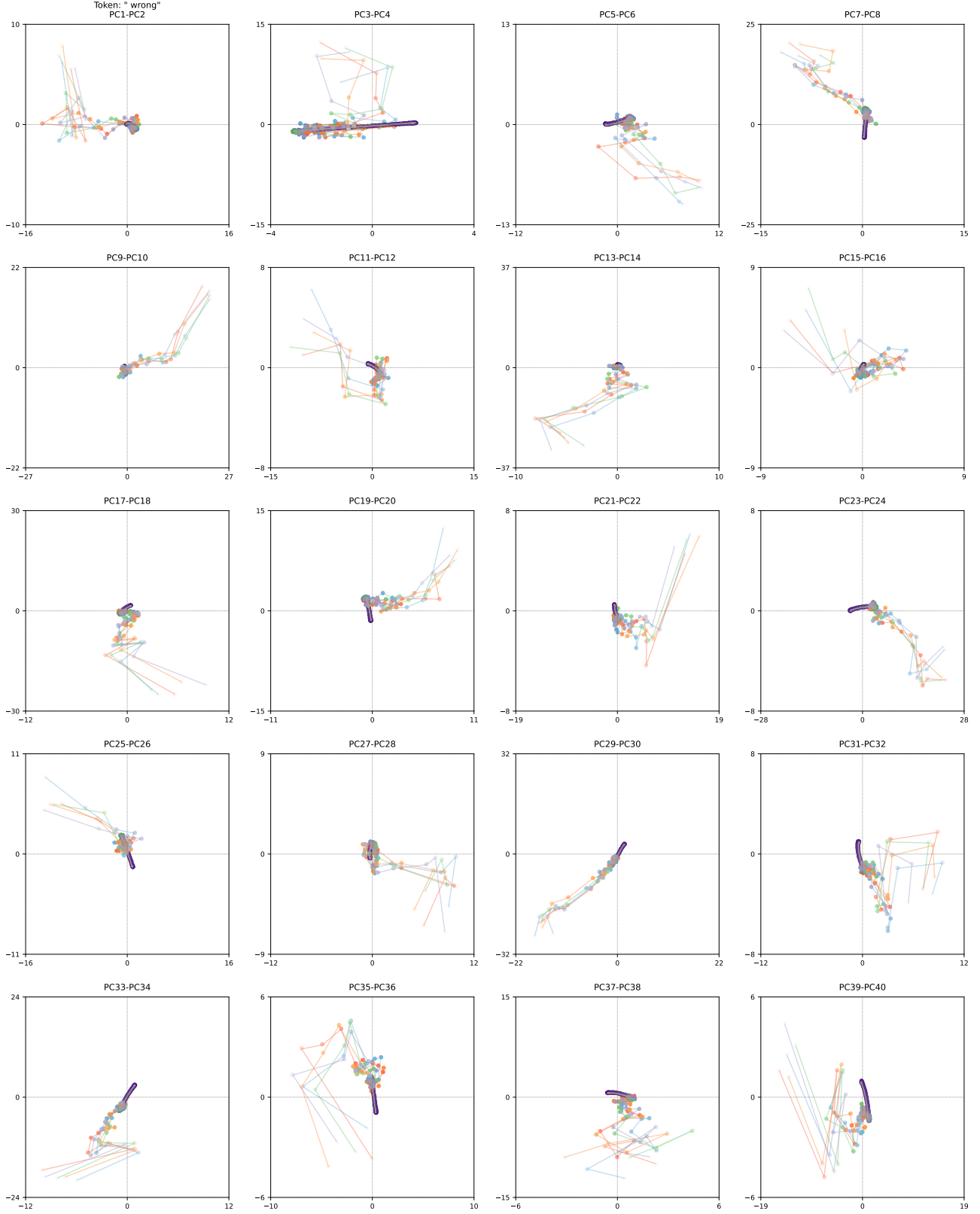


Figure 25: Detailed PCA of Latent Space trajectories for the unsafe question. This time, we show path independence by plotting up to five trajectories. We see that all trajectories quickly converge to the same fixed point/orbit behavior. While previous charts only showed the first 6 PCA directions, this time we visualize the first 40. Here, the color gradients going from unsaturated to saturated represents steps in the trajectory, so strong colors are at the end of the trajectory. Gray denotes the overlap of multiple trajectories.

C. Pretraining Data

Table 7: Datasets used for model pre-training (Part 1: Standard sources)

Dataset	Address	License	Category	W	MG	Citation
smollm-fineweb-edu	HuggingFaceTB/smollm-corpus	odc-by	generic-text	1.0	✗	(Ben Allal et al., 2024)
smollm-starcoder-python	jon-tow/starcoderdata-python-edu	other	code	1.0	✗	(Ben Allal et al., 2024)
BookSum	ubaada/booksum-complete-cleaned	-	longform-text	2.0	✗	(Kryściński et al., 2022)
GoodWiki	euirim/goodwiki	mit	longform-text	4.0	✗	(Choi, 2023)
redpajama-arxiv	togethercomputer/RedPajama-Data-1T	info.arxiv.org	scientific-text	2.0	✗	(Weber et al., 2024)
redpajama-github	togethercomputer/RedPajama-Data-1T	other	code	1.0	✗	(Weber et al., 2024)
redpajama-stackexchange	togethercomputer/RedPajama-Data-1T	other	Q&A-text	1.0	✗	(Weber et al., 2024)
dolma-CC-news	allenai/dolma	odc-by	generic-text	1.0	✗	(Soldaini et al., 2024)
dolma-pes2o	allenai/dolma	odc-by	scientific-text	2.0	✗	(Soldaini et al., 2024)
dolma-reddit	allenai/dolma	odc-by	generic-text	1.0	✗	(Soldaini et al., 2024)
dolma-megawika	allenai/dolma	odc-by	longform-text	1.0	✗	(Soldaini et al., 2024)
dolma-books	allenai/dolma	odc-by	longform-text	2.0	✗	(Soldaini et al., 2024)
dolma-wiki	allenai/dolma	odc-by	longform-text	4.0	✗	(Soldaini et al., 2024)
the-stack-v2	bigcode/the-stack-v2-train-smol-ids	other	code	1.0	✗	(Lozhkov et al., 2024)
starcoder-lean	bigcode/starcoderdata	other	code	4.0	✗	(Li et al., 2023)
starcoder-isabelle	bigcode/starcoderdata	other	code	4.0	✗	(Li et al., 2023)
starcoder-fortran	bigcode/starcoderdata	other	code	2.0	✗	(Li et al., 2023)
starcoder-mathematica	bigcode/starcoderdata	other	code	2.0	✗	(Li et al., 2023)
matrix-books	m-a-p/Matrix	apache-2.0	longform-text	0.25	✗	(Zhang et al., 2024a)
matrix-exams	m-a-p/Matrix	apache-2.0	Q&A-text	1.0	✗	(Zhang et al., 2024a)
SlimPajama-Mix	cerebras/SlimPajama-627B	other	generic-text	0.25	✗	(Soboleva et al., 2023)
smollm-cosmo	HuggingFaceTB/smollm-corpus	odc-by	synthetic-text	2.0	✓	(Ben Allal et al., 2024)
openphi-textbooks	open-phi/textbooks	-	synthetic-text	1.0	✓	(Colegrove et al., 2024)
openphi-textbooks-grounded	open-phi/textbooks_ground	-	synthetic-text	1.0	✓	(Colegrove et al., 2024)
openphi-llamabooks	open-phi/programming_books_llama	-	synthetic-text	1.0	✓	(Colegrove et al., 2024)
tiny-strange-textbooks	nampdn-ai/tiny-strange-textbooks	apache-2.0	synthetic-text	1.0	✓	(Nam Pham, 2024)
tiny-textbooks	nampdn-ai/tiny-textbooks	apache-2.0	synthetic-text	1.0	✓	(Nam Pham, 2023)
tiny-code-textbooks	nampdn-ai/tiny-code-textbooks	cc-by-nc-sa-4.0	synthetic-text	1.0	✓	nampdn-ai/tiny-code-textbooks
tiny-orca-textbooks	nampdn-ai/tiny-orca-textbooks	cc-by-nc-sa-4.0	synthetic-text	1.0	✓	nampdn-ai/tiny-orca-textbooks
sciphi-textbooks	SciPhi/textbooks-are-all-you-need-lite	llama2	synthetic-text	1.0	✓	SciPhi/textbooks-are-all-you-need-lite
textbook-programming	vikp/textbook_quality_programming	-	synthetic-text	1.0	✓	vikp/textbook_quality_programming
proofpile-algebra	EleutherAI/proof-pile-2	-	math	1.0	✗	(Azerbayev et al., 2023)
openweb-math	open-web-math/open-web-math	-	math	1.0	✗	(Paster et al., 2023)
british-library-books	biglam/blbooks-parquet	cc0-1.0	longform-text	1.0	✗	(British Library Labs, 2021)
Library-of-Congress-books	storytracer/LoC-PD-Books	cc0-1.0	longform-text	1.0	✗	(Majstorovic, 2024)
MathPile	GAIR/MathPile	cc-by-nc-sa-4.0	math	2.0	✗	(Wang et al., 2024a)
CLRS	tomg-group-umd/CLRS-Text-train	Apache-2.0	math	1.0	✓	(Markeeva et al., 2024)
AutoMathText-1	math-ai/AutoMathText	CC BY-SA 4.0	math	1.0	✗	(Zhang et al., 2024c)
AutoMathText-2	math-ai/AutoMathText	CC BY-SA 4.0	math	1.0	✗	(Zhang et al., 2024c)
AutoMathText-3	math-ai/AutoMathText	CC BY-SA 4.0	math	1.0	✗	(Zhang et al., 2024c)
bigcode-commitpack	bigcode/commitpackft	mit	code	1.0	✗	(Muennighoff et al., 2024)
bigcode-stack-python-fns	bigcode/stack-dedup-python-fns	other	code	1.0	✗	(Muennighoff et al., 2024)
VikpPython	vikp/python_code_instructions_filtered	-	code	1.0	✓	vikp/python_code_instructions_filtered
chessllm	mlabonne/chessllm	-	misc-reasoning	1.0	✗	mlabonne/chessllm
WaterHorseChess-pre	Waterhorse/chess_data	apache-2.0	misc-reasoning	1.0	✗	(Feng et al., 2023)
eleutherai-lichess	EleutherAI/lichess-puzzles	CC0 1.0	misc-reasoning	1.0	✗	(Schwarzschild et al., 2021a)